Assignment 1

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Introduction

This Assignment will cover four research questions listed below:

> ***RQ1****: Is there a relationship between GDP/capita and the number of personal cell phones per 100 individuals measured in 2005?*
>
> ***RQ2:*** *Has the electricity consumption per capita in Saudi Arabia increased from 1990 to 2005?*
>
> ***RQ3****: Is there a difference in income between the New York districts, Manhattan and Brooklyn?*
>
> ***RQ4****: Are there differences in house pricing (SEK/sqmr) in Sweden between 2021 and 2023?*

To answer each of the research questions, the program RStudio will be used accordingly.

Methods

This section will cover the methodology performed to answer the research questions posed in the introduction. Thus, this section is composed of four different parts, each of which refers to its respective research question.

The data utilized for this assignment was composed of four Excel documents (datasets) and one text (txt) document, all provided by the course instructors via Canvas. As instructed by the course instructors, RStudio was the main program used to run the statistical analyses on all four parts, whilst Excel was used to facilitate the combination of the datasets.

To support the data analyses, I used different online sources, including articles. Similarly, online cheatsheets and online forums were used to enhance the results. All these sources have been cited in the Reference section.

Part 1

The results of the below code lines is presented in suitable Tables or Figures in the Results section for later analysis in the Discussion section.

To answer the first research question: *Is there a relationship between GDP/capita and the number of personal cell phones per 100 individuals measured in 2005?* I used the data set named I ran the following Rstudio code to find the basic measures of tendency, mean, median, and standard deviation to get a better understanding of the data. Similarly, I also created two histograms of my two variables *Cellphones per 100* and *GDP per capita*. This way, I could find if the data I was working with was normally distributed.

```
mean(df$GDPperCAPITA)

mean(df$Cell_per_100)

median(df$GDPperCAPITA)

median(df$Cell_per_100)

sd(df$GDPperCAPITA)

sd(df$Cell_per_100)


hist(df$Cell_per_100, main = "Histogram of Cellphones",
     xlab = "Number of Cellphones",
     col = "blue",
     border = "black"
     )

hist(df$GDPperCAPITA,
     main = "Histogram of GDP per Capita",
     xlab = "GDP per Capita",
     col = "yellow",
     border = "black"
     )
```

Given that neither variables were normally distributed, and given the presence of outliers (see Results Part 1, *Figure 1*), I ran a Spearman Rank Coefficient to find the correlation between GDP per capita and the number of cell phones people have had using the following code:

```
cor.test(df$GDPperCAPITA, df$Cell_per_100, method = "spearman")
```

To get a better visualization of the data, I created a scatter plot. However, I noticed that the scatter plot was looking very clustered making it very difficult to interpret the relationship between the two variables, so I rewrote the code to fit the data into a scatter plot using a logarithmic function - see the code below.

```
ggplot(df, aes(x = GDPperCAPITA, y = Cell_per_100)) +
  geom_point(color="orange") +
  geom_smooth(method = "lm", color = "blue", se = FALSE) +
  labs(
    x = "GDP per Capita",
    y = "Cellphones per 100",
    title = "GDP per capita and Cellphones in 2005"
  )

#As the scatter plot was looking very clustered, I ran it again with LOG function

ggplot(df, aes(x = log(GDPperCAPITA), y = log(Cell_per_100))) +
  geom_point(color = "orange") +
  geom_smooth(method = "lm", color = "lightblue", se = FALSE) +
  labs(
    x = "Log of GDP per Capita",
    y = "Log of Cellphones per 100",
    title = "GDP per Capita and Cellphones in 2005"
  )
```

Once I could better visualize my data, the results were analyzed and discussed in the Discussion section with the main objective being answering my RQ1: *Is there a relationship between GDP/capita and the number of personal cell phones per 100 individuals measured in 2005?*

Part 2

To answer my second research question: *Has the electricity consumption per capita in Saudi Arabia increased from 1990 to 2005?* I used the dataset *Electricity_use_per_person.xlsx*. The best approach would be to run a linear regression analysis that would then be turned into a scatter plot, that way we could better understand if electricity consumption has risen between 1990 and 2005. However, three different linear regression models were created in an attempt to find the best fit.

Firstly, I filtered the data so I only had the values for the period 1990 to 2005. Then, I ran a linear regression analysis between the independent variable *Year* and the dependent variable *Energy Use*, this first was simply called *Model*. Afterwards, I ran two more linear regression analyses: one using the logarithm of the dependent variable (*Log of Energy Use)*, and the second one using the squared root of the dependent variable *(Squared Root of Energy Use)*, named *LogModel* and *SquareModel* respectively. I then ran basic diagnostic plots and a basic scatter plot. The results were then put next to each other to facilitate the comparison between the three models: *Model, LogModel* and *SquareModel*. This is the code used:

```r
library(dplyr)
Electricity_2005 <- Electricity %>% filter(Year <= 2005)

    #Basic Model
model <- lm(Saudi_Arabia ~ Year, data = Electricity_2005)

par(mfrow = c(1, 1))
plot(model)

plot(Electricity_2005$Year, Electricity_2005$Saudi_Arabia,
     main = "Electricity Consumption in Saudi Arabia",
     xlab = "Year", ylab = "Electricity Consumption per capita", pch = 16,
     xlim = c(min(Electricity_2005$Year), 2005)
     )
abline(model, col = "pink", lwd = 2)

    #Log Model
Reg.Model <- lm(log10(Saudi_Arabia) ~ Year, data = Electricity_2005)

plot(Reg.Model)

plot(Electricity_2005$Year, sqrt(Electricity_2005$Saudi_Arabia),
     main = "Electricity Consumption in Saudi Arabia",
     xlab = "Year", ylab = "Log10 of Electricity Consumption per capita",
     pch = 16
     )
abline(Reg.Model, col = "pink", lwd = 2)

    #Squared root Model
Reg.Model_sqrt <- lm(sqrt(Saudi_Arabia) ~ Year, data = Electricity_2005)

plot(Reg.Model_sqrt)

plot(Electricity_2005$Year, sqrt(Electricity_2005$Saudi_Arabia),
     main = "Electricity Consumption in Saudi Arabia",
     xlab = "Year", ylab = "Square Root of Electricity Consumption per capita",
     pch = 16
     )
abline(Reg.Model_sqrt, col = "pink", lwd = 2)

summary(model)
summary(Reg.Model)
summary(Reg.Model_sqrt)
```

As with Part 1, these results are shown in the Results section for later analysis in the Discussion section. The discussion led to the answer of the RQ2: *Has the electricity consumption per capita in Saudi Arabia increased from 1990 to 2005?*

Part 3

To answer my third research question, I started by importing the data provided by Lander_HousingNew.xlsx. I created a histogram to see if the data I had on income was normally distributed to determine whether a t-test was plausible or if a Wilcoxon rank-sum test would be more fitting, in the case of non-normally distributed data.

```r
options(scipen = 999)

hist(df$Income/1000,
     breaks=8,
     main = "Distribution of Income",
     xlab = "Income (in thousands)",
     col = "pink",
     border = "black",
     cex.axis = 0.75
     )
```

Given that the data was not normally distributed, see *Figure 10*. I ran a Wilcoxon rank-sum test to find if the distribution of income in Manhattan and Brooklyn differ from each other. In addition, the code below shows that I further obtained measures of central tendency for income in both districts. I also decided to include the minimum and the maximum value to add to the analysis.

```r
wilcox.test(df$Income[df$Boro == "Manhattan"],
            df$Income[df$Boro == "Brooklyn"]
            )

mean(df$Income[df$Boro == "Manhattan"])
median(df$Income[df$Boro == "Manhattan"])
sd(df$Income[df$Boro == "Manhattan"])
min(df$Income[df$Boro == "Manhattan"])
max(df$Income[df$Boro == "Manhattan"])

mean(df$Income[df$Boro == "Brooklyn"])
median(df$Income[df$Boro == "Brooklyn"])
sd(df$Income[df$Boro == "Brooklyn"])
min(df$Income[df$Boro == "Brooklyn"])
max(df$Income[df$Boro == "Brooklyn"])
```

Finally, I created a boxplot for a better visualization of the income distribution in both districts with the y-axis showing income in thousands. Despite the box-plot of Brooklyn being slightly squeezed, I decided this was the best representation of the income differences between the Districts, since making Brooklyn's box-plot less squeezed (for example, using log) would be an inaccurate visualization.

```r
boxplot(Income / 1000 ~ Boro, data = df,
        main = "Income Distribution by District",
        xlab = "District",
        ylab = "Income (in thousands)",
        col = c("turquoise", "plum"),
        border = "black",
        ylim = c(0, 10000),
        cex.axis = 0.75,
        outline = TRUE
      )

grid(nx = NULL, ny = NULL,
     col = "gray",
     lty = "dotted"
     )
```

Part 4

To approach this question, a paired sample test of median or mean difference would be meaningful. However, it was important to find out if the housing prices in both 2021 and 2023 followed a normal distribution. For this, I created a histogram for each year using the following code:

```r
colnames(Housepricing) <- c("Month", "sek_sqrm_21", "sek_sqrm_23")

par(mfrow = c(1, 1))
hist(Housepricing$sek_sqrm_21,
     breaks=6,
     main = "Housing Price Distribution 2021",
     xlab = "Price per sqrm",
     col = "lavender",
     border = "black",
     cex.axis = 0.75
)
xfit<-seq(min(Housepricing$sek_sqrm_21),max(Housepricing$sek_sqrm_21),length=40)
yfit<-dnorm(xfit,mean=mean(Housepricing$sek_sqrm_21),sd=sd(Housepricing$sek_sqrm_21))
yfit <- yfit*diff(h$mids[1:2])*length(Housepricing$sek_sqrm_21)
lines(xfit, yfit, col="orchid", lwd=2)

hist(Housepricing$sek_sqrm_23,
     breaks=6,
     main = "Housing Price Distribution 2023",
     xlab = "Price per sqrm",
     col = "wheat",
     border = "black",
     cex.axis = 0.75
)
xfit<-seq(min(Housepricing$sek_sqrm_23),max(Housepricing$sek_sqrm_23),length=40)
yfit<-dnorm(xfit,mean=mean(Housepricing$sek_sqrm_23),sd=sd(Housepricing$sek_sqrm_23))
yfit <- yfit*diff(h$mids[1:2])*length(Housepricing$sek_sqrm_23)
lines(xfit, yfit, col="orange", lwd=2)
```

Given that the data is not normally distributed (see *Figure 12* and *Figure 13*), I decided to run a paired Wilcoxon signed rank-test to find if there is any significant difference in the median prices in 2021, and in 2023. I used the paired Wilcoxon signed rank-test as the data sample is small, it does not follow a normal distribution, and our data is paired - we have the data paired by month, and we are looking at two years.

```r
wilcox.test(Housepricing$sek_sqrm_21,
            Housepricing$sek_sqrm_23,
            paired = TRUE,
            alternative = "two.sided"
            )
```

Finally, for better visualization and interpretation, I created a boxplot for the housing prices on each year.

```
attach(Housepricing)
boxplot(sek_sqrm_21,sek_sqrm_23,
        names = c("2021", "2023"),
        main = "Housing Prices in Sweden",
        ylab = "Price (SEK/sqmr)",
        xlab = "Year",
        col = c("royalblue2", "slateblue3"),
        border = "black",
        cex.axis = 0.75,
        outline = TRUE
)
grid(nx = NULL, ny = NULL,
     col = rgb(169/255, 169/255, 169/255, alpha = 0.5),
     lty = "dotted"
)
detach(Housepricing)
```

Results

For structure convenience, the results are divided into four parts similar to the previous section, one for each research question.

Part 1

Table 1. Measures of Central Tendency		
	GDP per Capita	Cellphones per 100
Mean	8’304.541	44.30079
Median	2’284.187	37.95
Standard Deviation	13’307.2	36.14692

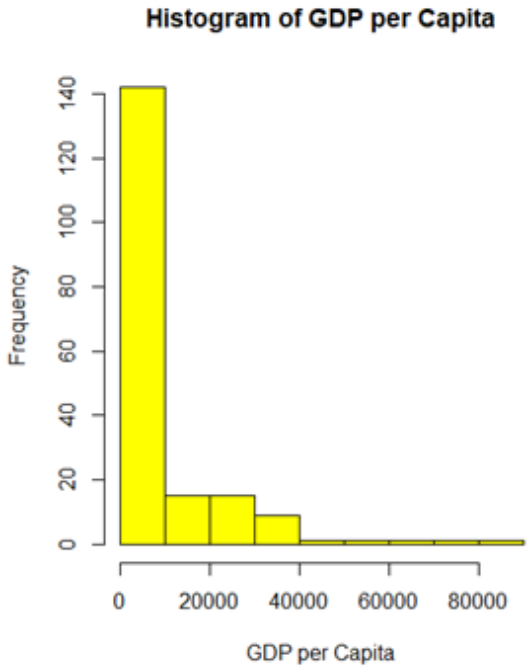


Figure 1. Histogram of GDP per capita 2005

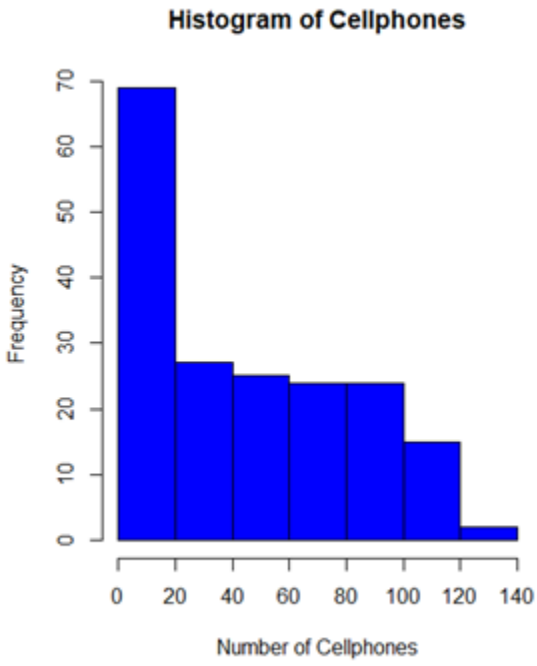


Figure 2. Histogram of Cellphones 2005

Figure 1 and *Figure 2* above show that *GDP per capita* nor *Cellphones* are normally distributed.

Table 2. Results of Spearman Rank Correlation	
S	148291
p-value	<2.2e-16
Rho	0.8617259

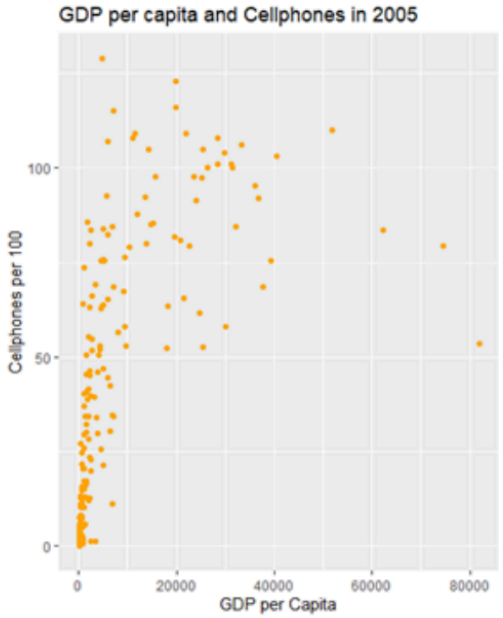


Figure 3. GDP per capita and Cellphones 2005

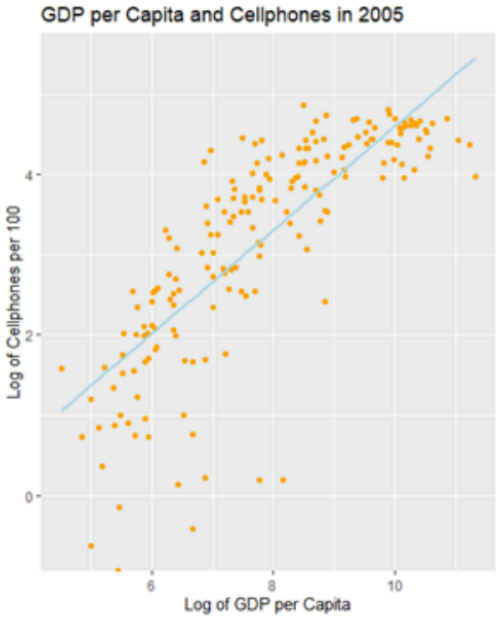


Figure 4. Log of GDP per capita and Cellphones 2005

Figure 3 was the first result of the scatter plot, and is clearly difficult to read and interpret. Therefore, the logarithmic scatter plot was run, resulting in *Figure 4*. *Figure 4* clearly shows a direct relationship between *GDP per capita* and *Cellphones*.

Part 2

Residuals vs. Fitted: Residuals seem to be randomly spread out for all models.

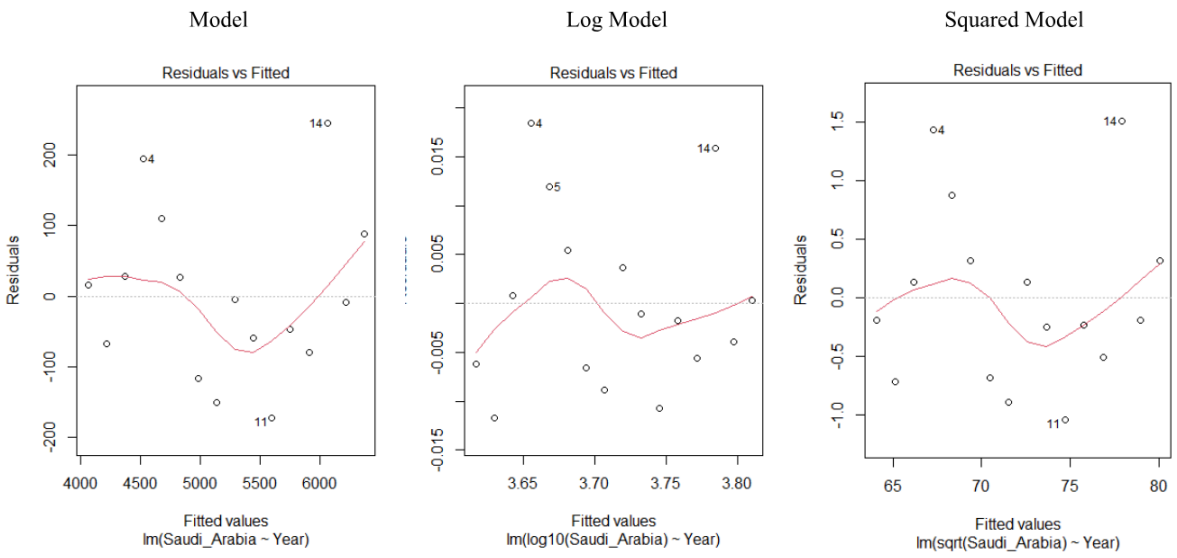


Figure 5. Residuals vs Fitted for Basic Model, Log Model and Squared Model

Normal Q-Q: The residuals in LogModel have the least deviation showing that they are normally distributed.

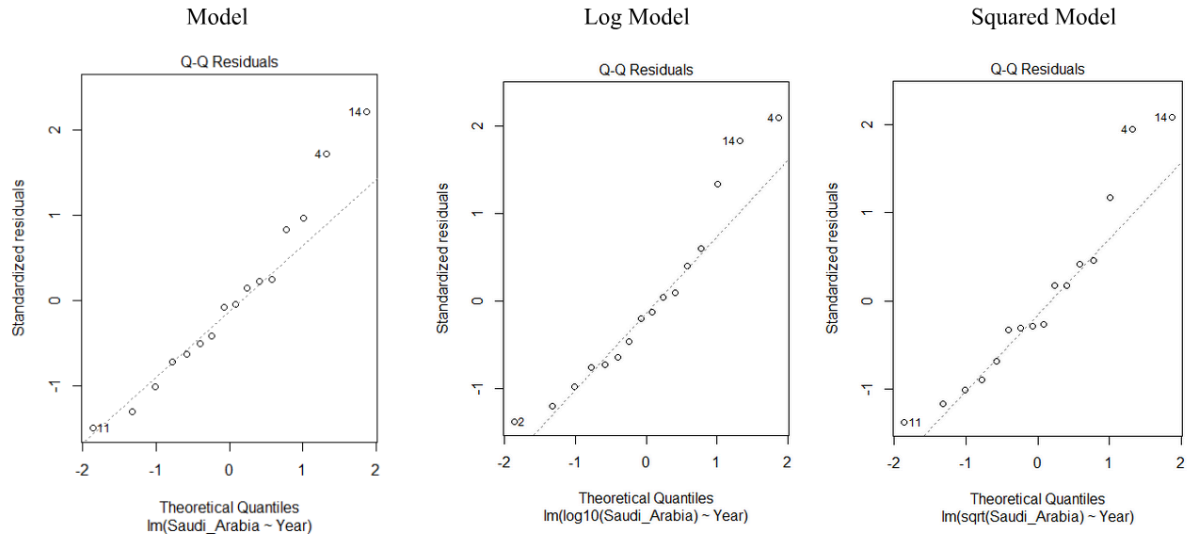


Figure 6. Q-Q Residuals for Basic Model, Log Model and Squared Model

Scale-Location: The three models show an equal variance (homoscedasticity) as the residuals seem randomly spread points.

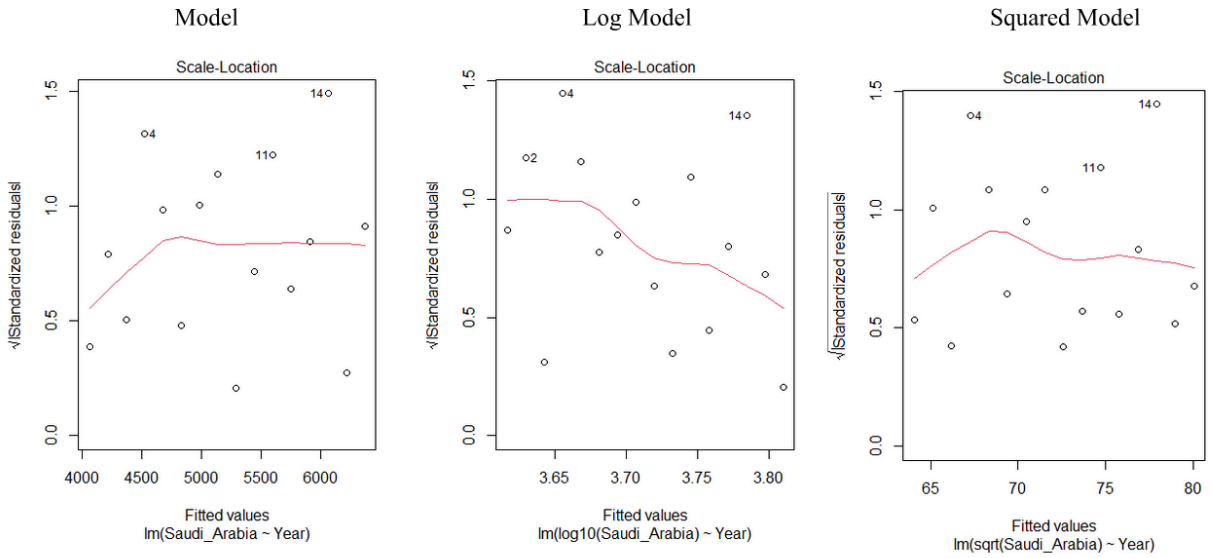


Figure 7. Scale-Location for Basic Model, Log Model and Squared Model

Residuals vs. Leverage: None of the models show a value that could influence the results of the linear regression.

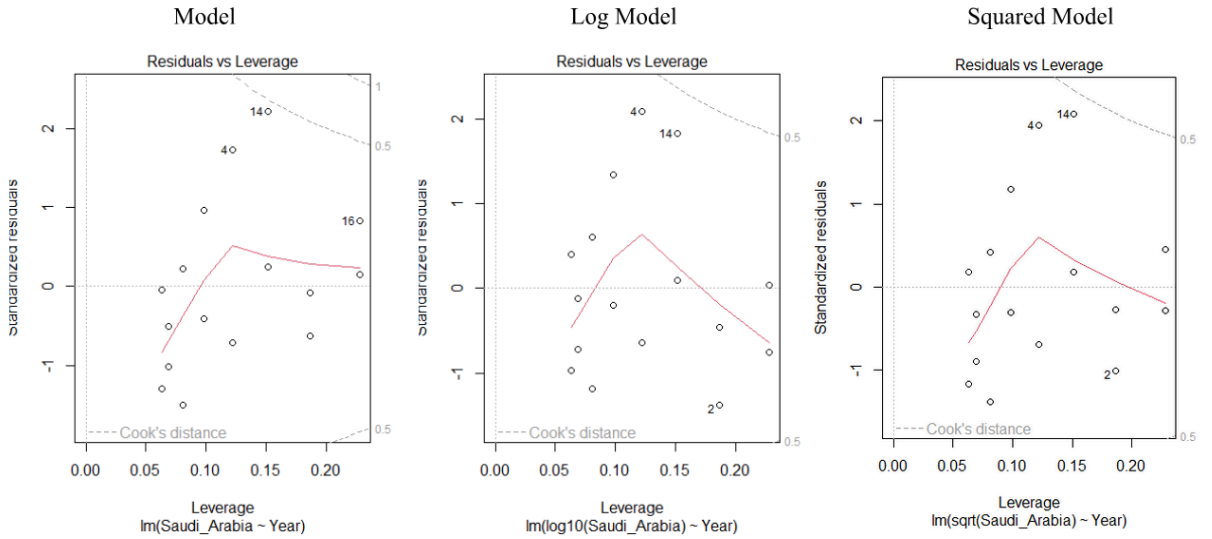


Figure 8. Residuals vs Leverage for Basic Model, Log Model and Squared Model

Scatter plots - LogModel did not allow me to add a line for best fit. The only difference is the scale in the Y-axis.

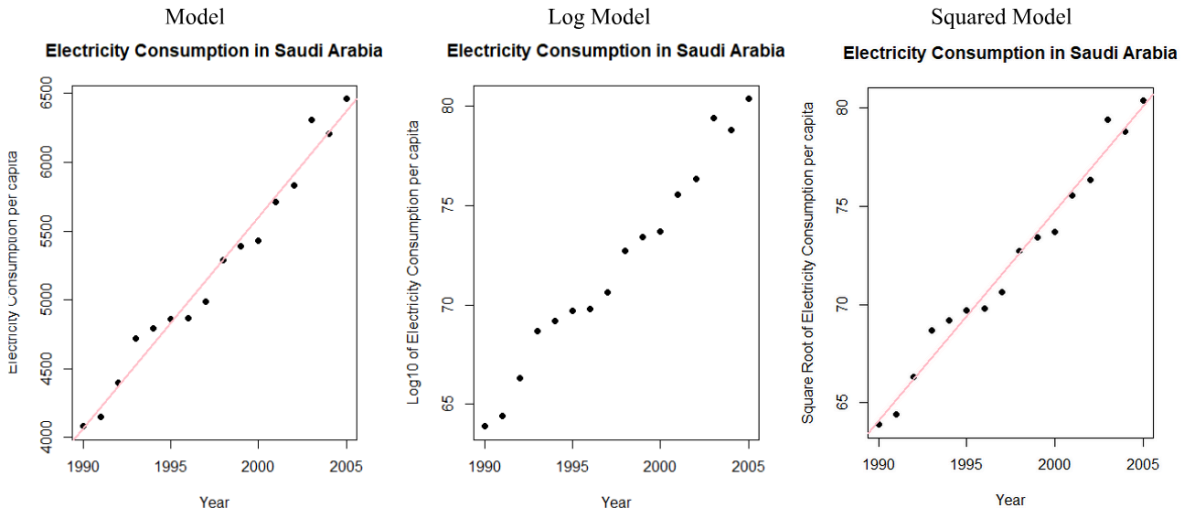


Figure 9. Scatter Plots for Basic Model, Log Model and Squared Model

Table 3. Summary of Regression for Model, LogModel and SquaredModel			
	Model	LogModel	SquaredModel
Residual Standard Error	120.4	0.009426	0.7852
Degrees of Freedom	14	14	14
R squared	0.9754	0.9784	0.9781
Adjusted R squared	0.9736	0.9768	0.9766
Regression coefficient *b*	153.9	0.01287	1.066
Standard Error	6.531	0.0005112	0.04258
t	-23.16	-22.54	25.03
p	1.156e-12	4.667e-13	5.038e-13

Part 3

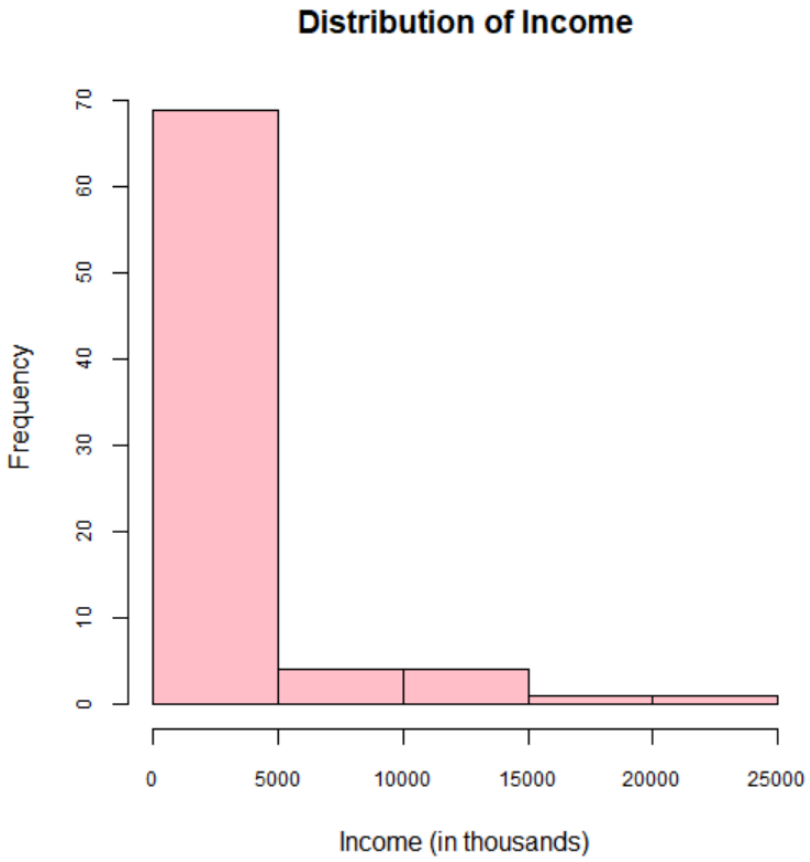


Figure 10. Distribution of Income

Table 4. Income in Manhattan and Income in Brooklyn - Wilcoxon rank-sum results	
Null Hypothesis: The income distribution of Manhattan and Brooklyn is the same	
w-value	1031
p-value	0.00001019
n	79

The p-value is very small, indicating strong evidence to reject the null hypothesis. This means that there is a statistically significant difference between the income distributions in Manhattan and Brooklyn.

Table 5. Measures of Central Tendency of Income		
	Brooklyn	Manhattan
Mean	639’548.1	3’344’961

Median	443’850.5	1’742’515
Standard Deviation	597’828.4	4’345’457
Min	147’206	147’229
Max	2’338’189	22’673’513

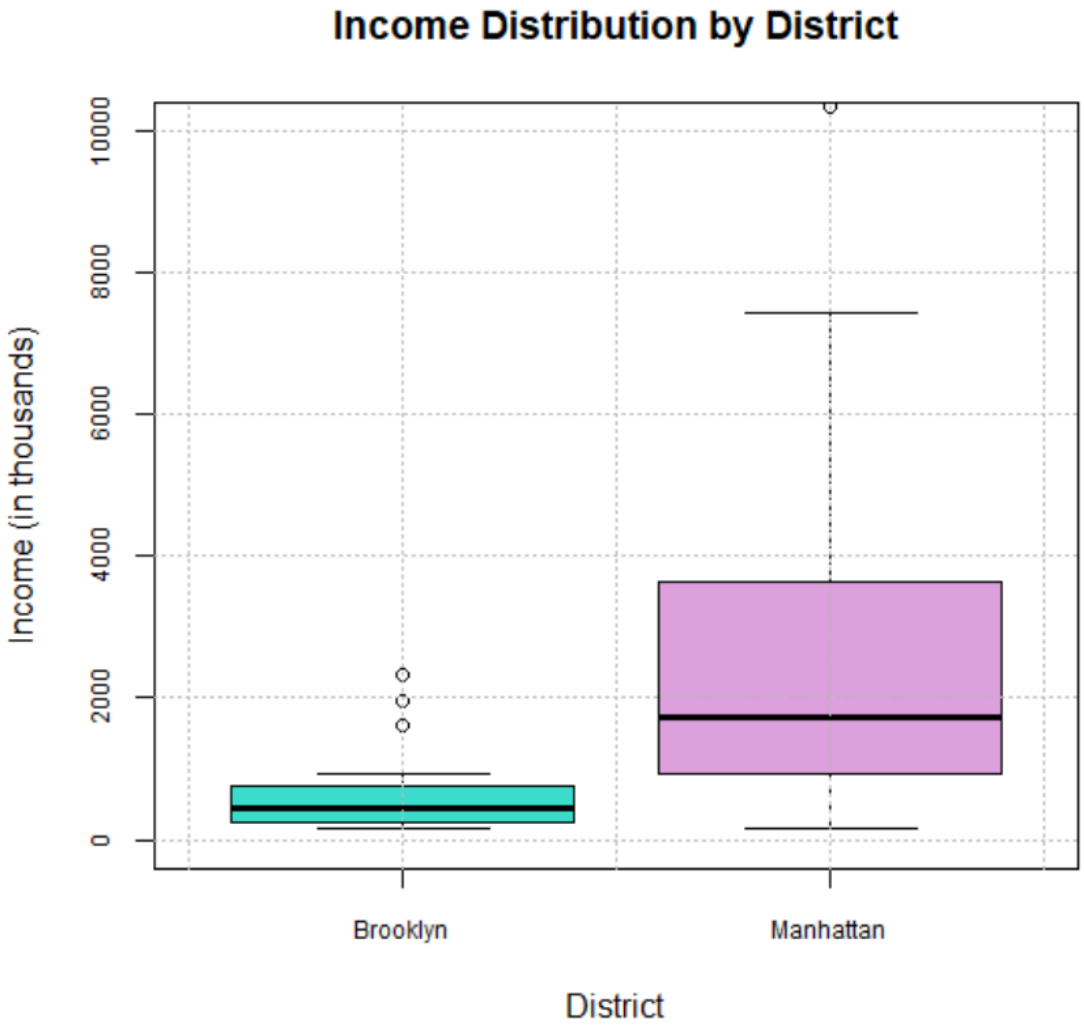


Figure 11. Box-plot showing the income distribution by district

Part 4

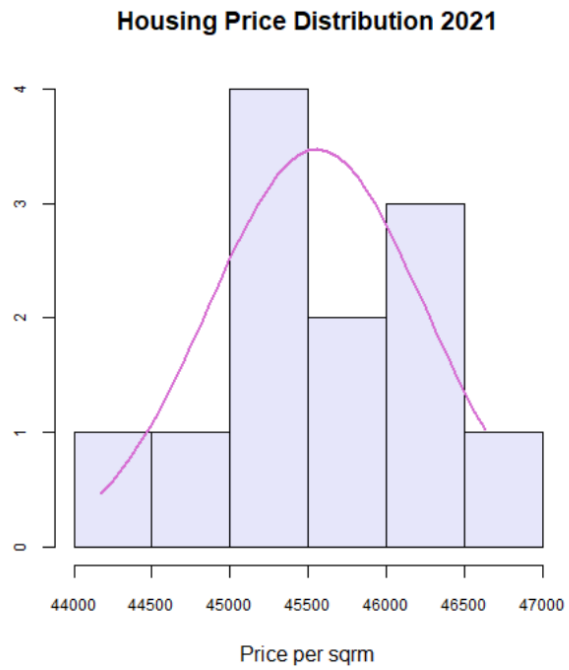


Figure 12. Histogram showing distribution of housing prices in 2021

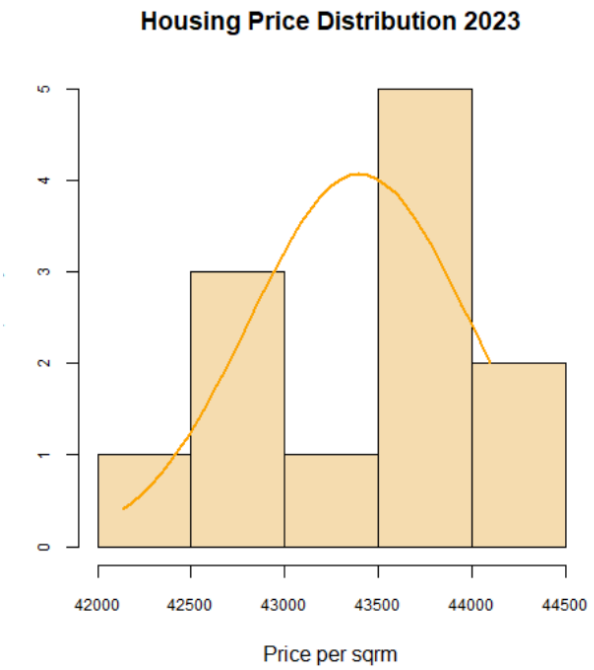


Figure 13. Histogram showing distribution of housing prices in 2023

Figure 12 and Figure 13 suggest that the housing prices in Sweden do not follow a normal distribution in 2021, nor 2023.

Table 6. Paired Sample Wilcoxon signed rank-test	
Null Hypothesis: The housing prices in Sweden in 2021 do not differ from those in 2023	
V-value	78
p-value	0.0004883
n	24



Figure 13. Housing Prices in Sweden (SEK/sqmr) in 2021 and 2023

Discussion

Part 1. Is there a relationship between GDP/capita and the number of personal cell phones per 100 individuals measured in 2005?

There seems to be a relationship between GDP per capita and the number of personal cellphones per 100 individuals in 2005. The results shown in the Results section Part 1 show that people in countries with higher GDP per capita tend to have more personal cell phones. The data consisted of more countries with lower GDP per capita as shown by *Figure 2*. Similarly, the frequency of the number of personal cell phones was higher in the lower numbers. *Table 1* made the interpretation more difficult, as the average GDP per capita was around 8'000 USD. This number, however, was influenced by those countries whose GDP per capita was incredibly bigger (what we could consider outliers), increasing the average. Therefore, *Figure 2* allows us to understand that most countries in the data had a GDP of 0 to 10'000 USD, and that the distribution of GDP per capita matched the distribution of cell phones shown in *Figure 1*.

Spearman rank's correlation test gave a value of 0.86. This clearly shows a strong direct relation between GDP per capita and cell phones per 100, as it is a positive value close to 1.00. *Figure 4* clearly shows a direct relationship between *GDP per capita* and *Cellphones*. Ultimately, we can see the relationship between the two variables more clearly in *Figure 4*. *Figure 4* gives a great insight into the further a country moves along the GDP per capita-axis the higher the number of cell phones per 100 people will be. Of course, it is important to highlight that there are countries that do not meet the relationship, as shown by the orange dots in *Figure 4* that are further away from the line of best fit. However, it has been shown that there is a moderately strong direct relationship between GDP per capita and the number of personal cellphones per 100 individuals in 2005.

Part 2. Has the electricity consumption per capita in Saudi Arabia increased from 1990 to 2005?

Firstly, the model that fits best to answer this question is the *LogModel*. Firstly, there was little variance in the diagnostic plots to determine which of the three models would be the best fit.

However, *LogModel* has the highest R-squared and adjusted R-squared values as shown in *Table 2*. This means that this model explains 98% of the variance in the analysis. Secondly, the *LogModel* has the lowest standard error by far, suggesting this model is more precise than, for example, the *Basic Model* with a standard error of 6.5. The scatter plot of *LogModel* does provide a good visualization of the data as we can see a clear point per year from 1990 to 2005.

Given the results in Part 2, and using the *LogModel*, we can say that the electricity consumption per capita has increased in Saudi Arabia from 1990 to 2005. As shown by the scatter plot of three models in *Figure 9*, there is an increase in energy consumption per capita as time progresses. Further, the variance in energy consumption can also be explained by time, as suggested by the R-values shown. The coefficient of *LogModel* is 0.01287, indicating that the logarithm of energy consumption per capita increases by 0.01287 each year. In other words, this corresponds to an approximate annual increase of 1.03 units in energy consumption per capita. Therefore, it can be said that the electricity consumption per capita in Saudi Arabia increased between 1990 and 2005.

Part 3. Is there any difference in income between the New York districts, Manhattan and Brooklyn?

The Results of Part 3 show that there is a big difference in income between the districts of Manhattan and Brooklyn. Firstly, we can see that from the Wilcoxon rank-sum results, the income distribution in both districts was not the same, given that the p-value was 0.00001019. The measures of central tendency in *Table 5* give us a better understanding of the differences between the two districts. The mean income in Brooklyn was around 600’000, whereas in Manhattan it was around 3 million. Similarly, the median income in Brooklyn was 400’000, whereas in Manhattan it was almost 2 million. This suggests a big difference in income earnings, with Manhattan’s income being much higher than Brooklyn’s. Further, we can also see that the minimum measured income in both districts is pretty much the same: 147’000. However, if we look at the maximum measured income, Brooklyn has a value of 2’338 million, and Manhattan has an outstanding 22’673 million.

Figure 11, further shows the great difference in income between Manhattan and Brooklyn. The median income of Manhattan lies outside of Brooklyn’s boxplot, pointing to a difference in

incomes. Brooklyn's box-plot indicates low data dispersion, meaning income levels show little variation. On the other hand, Manhattan's box-plot indicates higher data dispersion and a wider distribution, as its whiskers cover a much larger range than that of Brooklyn. Finally, we can see that Manhattan has outliers with significantly higher values than those in Brooklyn. From *Figure 11*, we can further see the skewness in the data on both districts, as there is a higher concentration in the first quartile.

Ultimately, from the data presented and the analysis provided, we can say that there is a big difference in income between the New York districts of Manhattan and Brooklyn, as Manhattan's mean and maximum incomes are considerably higher than those of Brooklyn.

Part 4. Are there any differences in housing prices in house pricing (SEK/sqmr) in Sweden between 2021 and 2023?

Given the results shown in *Part 4*, we can say that there has been notable changes in the housing prices in Sweden between 2021 and 2023. Firstly, the p-value of the paired Wilcoxon signed rank test indicates a significant statistical difference between the prices in 2021, and those prices in 2023 for each month, as p-value<0.05. The lower p-value suggests we can discard the null hypothesis, indicating disparities in the prices between the two years. It is possible to visually see the difference in prices in *Figure 13*.

In *Figure 13*, we can observe that the median price dropped from approximately 45'500 SEK/sqmr in 2021 to 43'500 SEK/sqmr in 2023, showing a significant decrease. Further, the range of prices shifted, with the 2023 box-plot moved down the y-axis towards lower prices. We can also see that the minimum price observed in 2021 became the maximum price in 2023, as shown by the whiskers in both boxes.

The price distribution of prices also changed. In 2021, most prices within the interquartile range (IQR) tended towards higher prices, whilst in 2023, we find a higher concentration of lower prices towards the lower end of the IQR.

Thus, the data shows a substantial decline in housing prices in Sweden from 2021 to 2023 - there is a difference in the housing prices between 2021 and 2023.

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